

HARSHIT JINDAL

Software Engineer

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EDUCATION

Chandigarh, India **University Institute of Engineering and Technology, Panjab University** **Jul 2017 – Jun 2021**
• Bachelor of Engineering (B.E.) in Information Technology - CGPA: 8.2/10 - <https://harshitjindal.com/graduate>

EXPERIENCE

Software Engineer **American Express** **Aug 2021 – Present**
• Working on backend API development using Flask to enable new features and enhance existing processes around the credit and fraud ML model builds and deployments.
• Parallely, working on migrating the existing codebase to a serverless architecture on the in-house cloud platform.
• Built 2 python modules and several standalone utilities to aid the Decision Science teams in developing and upgrading the ML models efficiently.

Analyst, Intern **American Express** **Jan 2021 – Jun 2021**
• Programmed 6 manual processes into end-to-end automated pipelines using Selenium, Pandas, Requests and Flask.
• Coded 100+ rules for the Dynamic Authorization of Credit (DAC) system that utilizes scores from in-house ML models to take the final decisions.

iOS Developer, Intern **C.S. Soft Solutions Pvt. Ltd.** **May 2019 – Jul 2019**
• Augmented Reality Home Planner: Used Apple ARKit to develop an app for a furniture store enabling users to place virtual 3D models of various items in their space to see how they would look and make better decisions.
• Food Delivery App: Contributed toward the development of a food delivery app based on a user story by enabling GPS tracking using Core Location, in addition to finding and displaying the optimal route between pickup and drop locations using the Directions API.

PROJECTS

- **[Computer Vision] Cosmic Artifact Filter:** Trained a deep learning model to flag cosmic ray artifacts in satellite imagery for deeper analysis using image data from NASA's MESSENGER Spacecraft orbiting the planet Mercury.
- **[Computer Vision] Dark Matter Discovery:** Built a pipeline to visualize and confirm the presence of dark matter in cosmic data based on the principle of Strong Gravitational Lensing, using PyAutoLens.
- **[Computer Vision] Accident Detection:** Used transfer learning to train a model capable of processing live video camera feed at traffic intersections to spot road accidents. Secured 3rd place at OctaHacks Hackathon 2019.
- **[Brain-Computer Interfacing] Cognitive Neuroscience:** Wrote software to detect muscle artifacts in real-time EEG data from the NeuroSky MindWave headset, with the potential for upgraded brain mapping by attenuation of the recognized artifacts. It was tested on the Left Frontal Lobe with >95% accuracy.
- **[iOS Software Development] Price Aggregator for Resellers:** Developed an app for a client in Bristol, United States that uses the iPhone camera to scan product barcodes using AVFoundation and aggregate the price history from different sources using API calls, enabling the resellers to make an informed decision.

ACHIEVEMENTS

6-month Deep Learning Nanodegree **Udacity / Facebook AI** **Nov 2018 – May 2019**
• Credential ID: C9HLTFDN - <https://harshitjindal.com/dl-nanodegree>
• Full Scholarship by Facebook (acceptance rate of 3%) - out of 10,000 global applicants of diverse qualifications.
• Coursework: Computer Vision, Recurrent Neural Networks, Generative Adversarial Networks, AWS Deployment.

SKILLS & TECHNOLOGIES

- C++, Python, Swift, JavaScript, Bash, SQL, Hive
- Flask, TensorFlow, PyTorch, Xcode, Git, Jenkins CI/CD, Docker with some exposure to Kubernetes
- Backend Web Development, iOS App Development, Machine Learning

VOLUNTEER & ADDITIONAL EXPERIENCE

- **Speaker at the Programming Club, UIET:** Guided freshmen CS students and taught introductory Machine Learning.
- **Mowgli Aid:** Volunteering for rescue and rehabilitation of stray and injured animals in the Chandigarh Tricity Area.
- **Udacity Mentor:** Facilitated smooth flow of Deep Learning Nanodegree program on Udacity's behalf.